



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year: 2023- 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CS3591 Computer Networks

Name of the Faculty member (s): Mrs.M.Swarna Sudha

Innovative Practice Description

Unit / Topic: Unit III / IP Addressing

Course Outcome: CO 4

Topic Learning Outcome: TLO 18

Activity Chosen: Learning by Teaching

Date of Implementation: 14.09.2023

Justification:

The significance of IP addresses lies in their fundamental role in identifying and locating devices within a network. Given their critical role, utilizing Learning by Teaching ensures a more effective learning experience for students. By actively involving students in the teaching process, they are not only compelled to understand the complexities of IP addresses but also to explain and articulate this essential concept to their peers. This active engagement raises a deeper level of understanding, making the information more memorable. Since IP addresses serve as the backbone of network communication.

Time Allotted for the Activity: 35 minutes

Details of the Implementation:

- Before the class, the teacher, inquired about the students' interest in teaching the specific topic Subhiksha V, Srivadhani Karthiga A, and Sundara Mahalingam P expressed interest.
- Based on their interest, Subhiksha V, Srivadhani Karthiga A, and Sundara Mahalingam were given topics on both IPV4 and IPV6.
- The adequately time was given to the students to prepare for their teaching role so the topic was given a day before the class.
- Students presented their allocated topics to their classmates
- To keep the class interested, students utilized a variety of instructional strategies. Subhiksha V employed real-world examples of IP addresses in use, such as websites, devices, or network configurations, to demonstrate the practical relevance of the concept Srivadhani Karthiga A used visual aids flashcards with visual elements to aid in quick recall of key IP address information, including subnet ranges and Sundara Mahalingam

P used animated videos that visually represent how IP addresses function, offering a dynamic and interactive way for students to grasp abstract concepts.

- To promote involvement, students were allowed to ask questions to their peers during the presentation. Because of this, students were able to have conversations and share their perspectives about topic which very important in Network concepts.

CO – PO / PSO mapping:

CO	PO1	PO2	PO4	PO9	PO10	PSO1
CO 3	2	2	1	1	1	1

(1 – Low 2 – Moderate 3 High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO4	PO9	PO10	PSO1
	2	2	1	1	1	1
Justification for correlation	Apply mathematical concepts to analyze and design IP Addressing solutions.	Develop the ability to identify engineering problems related to IP addressing.	Analyze the problem to find the solution for IP addresses issues	Encourage students to explore these modules independently to build a foundational understanding.	To Communicate effectively on complex engineering activities	Will be able to solve complex computation task

Images / Screenshot of the practice:



Figure 1: Subhiksha V , Srivadhani Karthiga and by Sundara Mahalingam P Teaching the Concepts of IP Addressing

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- All the students were eagerly listened the class. Students felt they are able to understand the topic.
 - The students were able to remember key details and concepts more easily
 - The students express that role of a teacher boost their confidence in knowledge of IP addresses
 - Students highlighted that this practice helped them improve their communication skills, such as clarity, organization, and the ability to present complex ideas in a simplified manner.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students actively engaging in teaching their peers gain a more profound understanding of the IP address
 - Actively participating in the teaching process helps students remember key details and concepts better compared to passive learning methods
 - This activity helped the student to improve communication skills.
 - Students spent their time in self-learning.
 - This activity encouraged the students to share their knowledge with others.

- ❖ **Challenges faced in implementation:**

- Effectively motivate students who are not participating in the activity due to shyness, fear of public speaking, or lack of confidence. Motivate the students to overcome their shyness and engage in active learning.
 - Implementing Learning by Teaching required additional time compared to traditional teaching methods. Planning and executing interactive teaching sessions was challenging.

References:

1. <https://effectiviology.com/protege-effect-learn-by-teaching>
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development
3. https://www.ritrjpm.ac.in/images/computer-science/2022-023/4.Collaborative_Coding_DBMS_KVS.pdf
4. <https://www.evergreen.edu/sites/default/files/facultydevelopment/docs/WhatisCollaborativeLearning.pdf>